Hologram VFX Documentation (Unity 6 URP)

By Sekochi Production

Last Update: 10 November 2025

Table of Contents

Introduction 2
Package Contents 2
Requirements 3
Installation 3
Usage 4
- Using the Hologram Material 4
- Creating a Hologram Material from Template Shader Graph 4
- Using the VFX Graph 4

Custom Properties 5
- Hologram Material 5
- VFX Graph 7

Troubleshooting 9
Contacts 9

Introduction

The Hologram VFX package demonstrates a modular hologram visual effect built using Shader Graph and VFX Graph in Unity 6 URP.

It is designed to work with any 3D model and provides adjustable parameters for designers and developers to achieve a variety of holographic looks.

This guide explains how to import, set up, and customize the effect step by step.

Package Contents

Within this project, contains the contents of the package. Here's an overview of the top hierarchy of the Package Contents:

Folder	Description
Demo/	Contains example scenes and scripts showing how to set up and use the Hologram VFX with different models and configurations.
Demo/Scripts/	Contains scripts specific for the demo scene.
Demo/Scene/	Contains the demo scene.
Materials/	Pre-configured materials that use the Hologram Shader Graph, ready to assign to your 3D models.
Models/	3D models used for demo purposes or testing the hologram effect.
Models/HologramProjector/	3D model used for demo purposes. Containing the Hologram Projector's Mesh and Texture.
Models/StanfordBuddha/	3D model used for demo purposes. Containing the Stanford Buddha's Mesh and Texture.
Models/StanfordDragon/	3D model used for demo purposes. Containing the Stanford Dragon's Mesh and Texture.
Models/Suzzane/	3D model used for demo purposes. Containing the Suzzane's Mesh and Texture.
Models/UtahTeapot/	3D model used for demo purposes. Containing the Utah Teapot's Mesh and Texture.

Prefabs/	Prefabricated GameObjects including hologram emitters, demo objects, and fully set-up VFX prefabs for quick use.
Shader Graphs/	Contains all Shader Graph files used to create the hologram material. Easily editable for customization.
Shader Graphs/Examples/	Contains the example Shader Graphs that demonstrates each functionality and applied in the demo scene.
Shader Graphs/Sub Graphs/	Contains the Sub Graphs that help build the Hologram's Shader Graph.
Textures/	Supporting textures for visual effects used in the hologram VFX.
VFX/	Unity VFX Graph files for the hologram projection ray and dust particle effects.
README.md	This file contains an outline of instructions, setup guide, and documentation for using the Hologram VFX package.

Requirements

- Unity 6.0 or newer
- Using the Universal Render Pipeline (URP)
- Shader Graph package installed
- VFX Graph package installed

Installation

- Clone or download the package from the Unity Asset Store or GitHub.
- Open your Unity 6 URP project.
- Import the Hologram VFX package into your Assets folder.
- (Optional) Ensure URP is properly configured:
    - Go to Project Settings > Graphics > Assign URP Asset

Usage

To use this effect, there are a few setups to configure into the object. Bla bla bla.

Using the Hologram Material

The following guide instructs how to use the hologram material:

1. Create a 3D object (e.g., Cube, Sphere, or custom model).
2. Locate the Hologram Material within the Materials folder.
3. Apply this Material by dragging the selected material to the model in the Scene view or apply it through the MeshRenderer component of the GameObject.
4. Tweak the material properties to customize the effect.

Creating a Hologram Material from Template Shader Graph

The following guide instructs how to create a new material from a template shader graph:

1. Locate the Hologram's Template Shader Graph within the Shader Graphs folder.
2. Duplicate the template to make a copy of it for easier organization.
3. Open the new duplicated template shader graph by double-clicking or right-click and select open.
4. Within the template shader graph, there are already some nodes being set up to create the Hologram Effect. You can tweak these nodes to customize. It is recommended to read some of the notes within the template to guide how each group of nodes work.
5. After customization, to apply this to an object, we need to create a material from the Shader Graph. We do this by right-clicking and navigate to Create > Material.

Using the VFX Graph

The following guide instructs how to use the VFX object:

1. Locate the VFX Graph object within the VFX folder.
2. To use this VFX object, drag the selected VFX object to the Scene and place it accordingly.
3. Tweak the VFX properties to customize the effect.

Custom Properties

To customize the Hologram Material and VFX Graph there are a handful of properties you can adjust to achieve the effect that you want.

Hologram Material

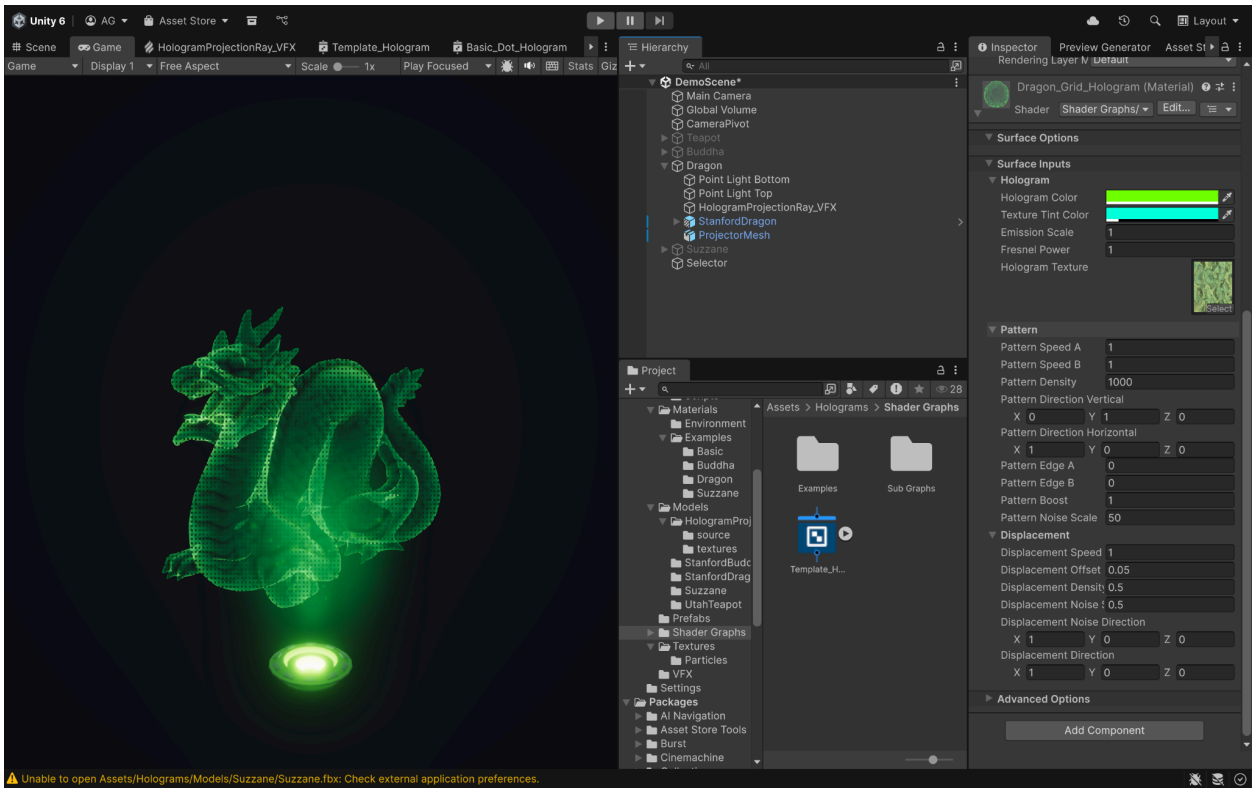

Figure 1: Customizing the Material Property

Property	Description
Hologram	
Hologram Color	The color of the hologram's pattern and rim light.
Texture Tint Color	The tint color applied to the object; interpolates with the object's texture if a hologram texture is set.
Emission Scale	Multiplier for the intensity of the hologram's pattern and rim light.
Fresnel Power	Controls how strongly the hologram edges glow depending on the viewing angle.
Pattern	
Pattern Speed (X)	Controls the speed at which the pattern moves along its

	specified direction.
Pattern Density	Adjusts the spacing or repetition frequency of the pattern across the surface.
Pattern Direction (Horizontal/Vertical/X)	Sets the direction in which the pattern travels. Controlled by a 3D Vector.
Pattern Edge (X)	Adjusts the smoothness or sharpness of the pattern edges using a smoothstep function.
Pattern Boost	Multiplies the emission intensity of the pattern.
Pattern Noise Scale	Controls the scale of the noise applied to the pattern, distorting its intensity for a more dynamic effect.
Displacement	
Displacement Speed	Controls how fast the displacement noise pattern of the hologram scrolls over time.
Displacement Offset	Sets the offset value applied to the vertices.
Displacement Density	Adjusts the frequency of vertex displacement across the surface.
Displacement Noise Scale	Controls the scale of the displacement noise pattern that distorts vertex positions for a glitchy effect.
Displacement Noise Direction	Defines the direction in which the displacement noise pattern scrolls to. Controlled by a 3D Vector.
Displacement Direction	Defines the direction in which the noise affects vertex displacement (along Axis, Direction, or Normal). Controlled by a 3D Vector.

VFX Graph

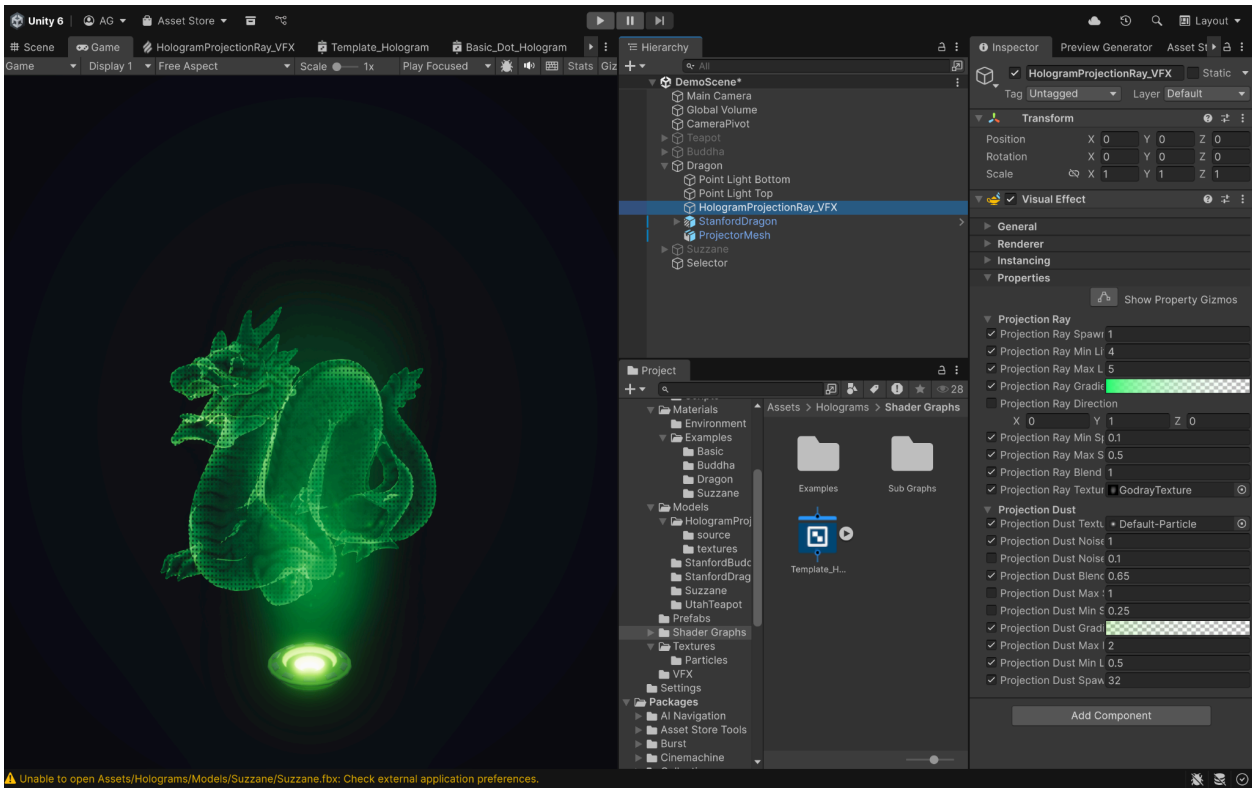

Figure 2: Customizing the VFX Property

Property	Description
Projection Ray	
Projection Ray Spawn Rate	Controls how many projection ray particles are spawned per second.
Projection Ray Min Lifetime	The minimum lifetime of each particle in seconds.
Projection Ray Max Lifetime	The maximum lifetime of each particle in seconds.
Projection Ray Gradient	Defines the color of particles over their lifetime.
Projection Ray Direction	The direction in which particles are emitted.
Projection Ray Min	The minimum speed of emitted particles.

Speed	
Projection Ray Max Speed	The maximum speed of emitted particles.
Projection Ray Blend	Controls how particle velocity blends with other motion or forces.
Projection Ray Texture	Texture applied to the particles for visual style (e.g., fog, dust, light streaks).
Dust	
Projection Dust Spawn Rate	Controls how many projection dust particles are spawned per second.
Projection Dust Min Lifetime	The minimum lifetime of each particle in seconds.
Projection Dust Max Lifetime	The maximum lifetime of each particle in seconds.
Projection Dust Gradient	Defines the color of particles over their lifetime.
Projection Dust Min Speed	The minimum speed of emitted particles.
Projection Dust Max Speed	The maximum speed of emitted particles.
Projection Dust Blend	Controls how particle velocity blends with other motion or forces.
Projection Dust Noise Intensity	Controls how strongly the noise affects the dust particles, creating variation in their movement or appearance.
Projection Dust Noise Frequency	Adjusts the frequency of the noise pattern applied to dust particles, affecting the scale of turbulence or randomness.
Projection Dust Texture	Texture applied to the particles for visual style (e.g., fog, dust, light streaks).

Troubleshooting

Below are some issues that could be encountered within the scope of this project. These are some of the troubleshooting approaches:

FAQ	Answer
Why does the material appear pink?	Ensure that URP and Shader Graph are properly installed. This package is designed for Unity's Universal Render Pipeline and may not work correctly on other pipelines.
Why is the VFX Graph not rendering?	Make sure all required packages (VFX Graph and URP) are installed and enabled in your project. Check that the scene camera can see the VFX emitter.
Why did my object disappear when applying the material?	Ensure the Hologram Color and/or Texture Tint Color alpha value is within the visible range (not fully transparent). Make sure the Displacement Offset is within a reasonable range; otherwise, vertices may be displaced out of view.

Contacts

- Developer Email: sekochi.dev@gmail.com
- Asset Store Page: [Unity Asset Store Link]